

PolicyImpact

Evidence-Grounded Agentic AI for Policy Change-to-Claim Review

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EXECUTIVE PREMISE Use language models to propose and challenge evidence-linked policy changes—not to calculate claim outcomes. Human-approved typed rules enter a deterministic engine; ambiguous language stays visible and non-executable.

1 What “evidence-grounded” means

Evidence-grounded policy change-to-claim review is a controlled workflow that links every proposed material change and rule field to exact old/new policy sections, records an independent critique, permits one bounded revision, requires an explicit human decision, and then applies only approved typed rules through deterministic operators. The output is a prioritized, auditable review queue—not a claim, payment, coverage, fraud, or clinical decision. Versioned evidence and named actors follow the same trust logic that HL7 assigns to provenance: identify the entities, activity, and agents that influenced an artifact (HL7 International, 2019).

2 Why this fits healthcare content management

Healthcare policy content is consequential because changes must be translated into repeatable operational logic while exceptions and uncertainty remain reviewable. CMS-0057-F reinforces the strategic direction toward computable exchange by requiring specified payer-facing FHIR APIs for affected programs, although this proof of concept neither performs prior authorization nor implements FHIR (CMS, 2024). ONC’s HTI-1 materials similarly emphasize source attributes, plain-language access, identified users, feedback, and risk management for covered decision-support interventions; these are useful transparency design signals rather than a statement that this demo is certified health IT (ONC, 2024).

Agentic AI can organize a sequence of role-bound proposals and checks, but unconstrained generation introduces unsupported content risk (Li et al., 2023). Healthcare evaluations also show that strong benchmark performance can coexist with clinically important gaps, motivating multidimensional human evaluation and cautious scope (Singhal et al., 2023). PolicyImpact therefore uses two stateless roles: a Policy Analyst proposes typed changes with evidence and qualifications; an Evidence Auditor accepts, rejects, revises, or abstains. The deterministic engine remains separate.

3 Architecture: language for interpretation, code for execution

POLICY PAIR	ANALYST	AUDITOR	HUMAN GATE	RULE ENGINE
Exact sections Controlled parser	Changes + rules Evidence links	Critique + one revision round	Approve / reject No silent action	Fixed operators Claim-impact queue

The design is intentionally smaller than a production stack: one controlled Markdown grammar, Pydantic schemas, a recorded offline role fixture with an optional live provider, and deterministic Python. No React application, vector database, fine-tuning, real codes, PHI, contracts, fee schedules, or autonomous tools are included.

4 Proof of concept and controlled evaluation

The synthetic scenario compares policy SYN-PAY-042 version 1.0 with version 2.0 across six sections and the same 16 invented claim records. Service date is retained as a field but does not choose a policy version; applying both versions to identical claims isolates the effect of policy content. Manual ground truth defines four material changes, three executable old/new rule pairs, one ambiguous abstention, expected evidence, and all 16 outcomes.

4 / 4 changes	6 / 6 typed rules	8 / 8 evidence keys	1 / 1 abstention	16 / 16 claim records
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The initial auditor accepted diagnosis and unit changes, requested a bounded clarification for modifier/documentation routing, and abstained on undefined transition wording. After revision, three rule pairs were accepted for human review and the ambiguity remained non-executable. The checked-in reviewed fixture then matched all specified records; unsupported assertions in the final audit were zero and traceability was 20/20. These perfect values are scenario-consistency checks for a fixture aligned to one synthetic ground truth—not live-model accuracy, generalization, savings, or production readiness.

5 Cotiviti opportunity, threat, and control

OPPORTUNITY	THREAT	BOUNDED CONTROL
Shorten policy-change triage and expose exact evidence.	Plausible but unsupported rule fields.	Typed schema, independent audit, abstention, assertion count.
Create a consistent handoff from content experts to rule teams.	Silent automation can hide accountability.	Explicit human approval; immutable decision log; deterministic execution.
Prioritize potentially affected records for expert review.	Scope drift into claim or care decisions.	Selected-criterion outcomes only; no medical necessity, denial, fraud, or payment result.

This boundary aligns with Cotiviti's public responsible-AI position: secure controlled environments, human experts responsible for decisions, and no AI medical-necessity determinations or denial of care (Cotiviti, n.d.). NIST's govern-map-measure-manage lifecycle and Generative AI Profile provide a practical governance frame for use-case definition, risk measurement, monitored deployment, and rollback (NIST, 2023, 2024).

6 Recommended next action: a controlled, result-blind pilot

- Select 10–20 approved historical policy pairs with no PHI in the prompt set; freeze versions, sections, and material-change criteria before model runs.
- Have independent policy/rule experts author a gold set, disagreements, and executable/non-executable boundaries without seeing model output.
- Run offline and live modes in a Cotiviti-controlled environment; measure change precision/recall, rule-field match, evidence accuracy, abstention, unsupported assertions, reviewer overrides, and traceability.
- Advance only if predefined gates pass. Retain a kill switch, access controls, versioned prompts/models, evidence logs, reviewer sign-off, and incident/rollback procedures. If e-PHI enters scope, conduct organization-specific privacy, security, and legal review; HHS risk-analysis guidance covers confidentiality, integrity, and availability across electronic media (HHS, n.d.).

BOTTOM LINE PolicyImpact demonstrates a governable handoff: generative roles surface and challenge evidence-linked candidates; humans authorize scope; deterministic code produces a review queue. The next claim worth testing is workflow reliability under result-blind, multi-policy evaluation—not production savings.

AI/software disclosure. OpenAI ChatGPT/Codex assisted with code and evidence auditing, literature organization, and drafting/editing. The author reviewed the analyses, citations, and text and is responsible for the content. Python, Pydantic, pytest, Streamlit, python-docx, and the presentation artifact tooling were used. No live model was invoked for the reported evaluation.

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